Solution: Bitonic Array Maximum

Problem Statement

Find the maximum value in a given Bitonic array. An array is considered bitonic if it is monotonically increasing and then monotonically decreasing. Monotonically increasing or decreasing means that for any index i in the array `arr[i] != arr[i+1]`.

Example 1:

```
Input: [1, 3, 8, 12, 4, 2]
Output: 12
Explanation: The maximum number in the input bitonic array is '12'.
```

Example 2:

```
Input: [3, 8, 3, 1]
Output: 8
```

Example 3:

```
Input: [1, 3, 8, 12]
Output: 12
```

Example 4:

```
Input: [10, 9, 8]
Output: 10
```

Solution

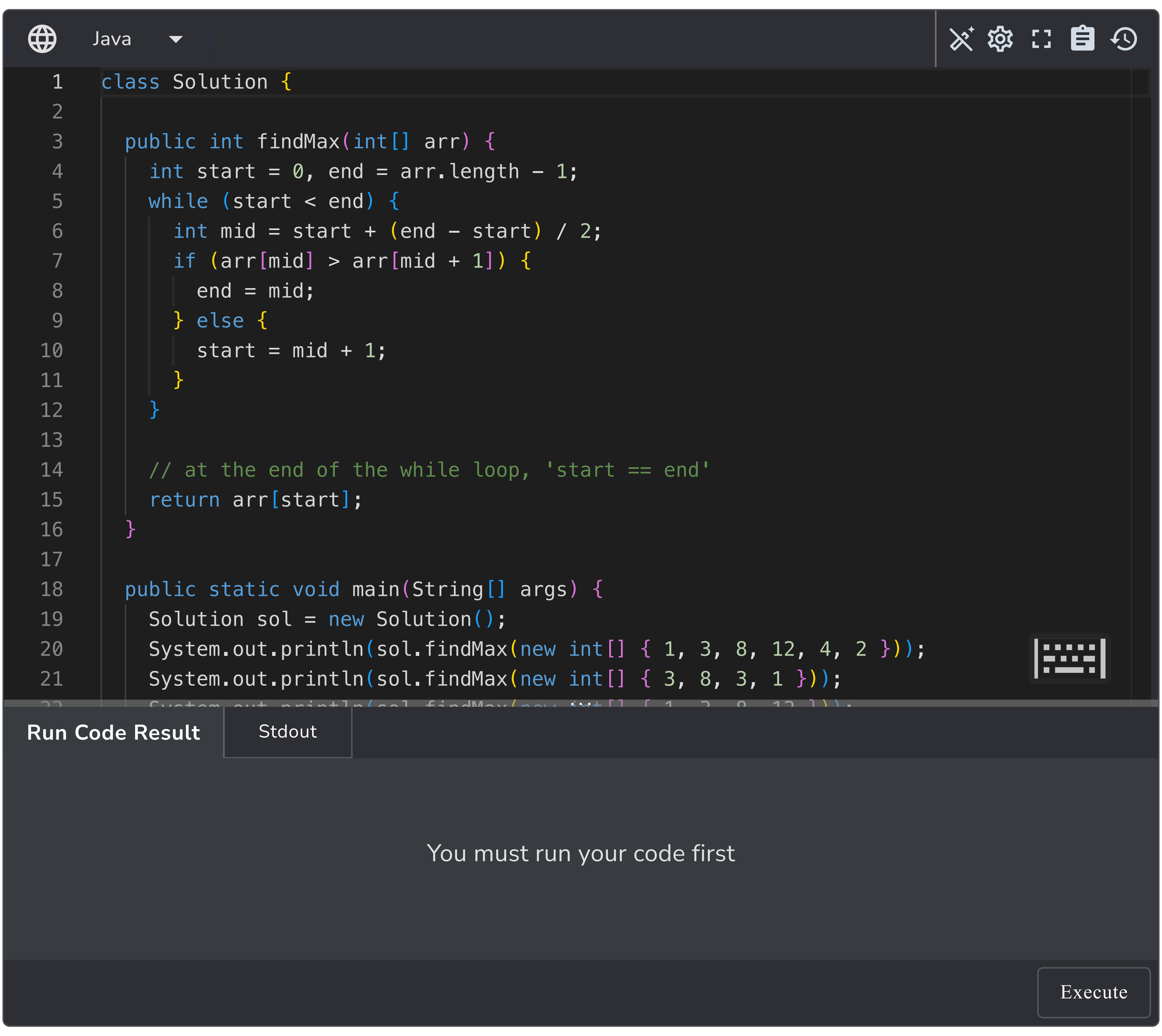
A bitonic array is a sorted array; the only difference is that its first part is sorted in ascending order and the second part is sorted in descending order. We can use a similar approach as discussed in **Order-agnostic Binary Search.** Since no two consecutive numbers are same (as the array is monotonically increasing or decreasing), whenever we calculate the `middle`, we can compare the numbers pointed out by the index `middle` and `middle+1` to find if we are in the ascending or the descending part. So:

1. If `arr[middle] > arr[middle + 1]`, we are in the second (descending) part of the bitonic array. Therefore, our required number could either be pointed out by `middle` or will be before `middle`. This means we will be doing: `end = middle`.
2. If `arr[middle] < arr[middle + 1]`, we are in the first (ascending) part of the bitonic array. Therefore, the required number will be after middle. This means we will be doing: `start = middle + 1`.

We can break when `start == end`. Due to the two points mentioned above, both `start` and `end` will be pointing at the maximum number of the bitonic array.

Code

Here is what our algorithm will look like:



```java
class Solution {

  public int findMax(int[] arr) {
    int start = 0, end = arr.length - 1;
    while (start < end) {
      int mid = start + (end - start) / 2;
      if (arr[mid] > arr[mid + 1]) {
        end = mid;
      } else {
        start = mid + 1;
      }
    }

    // at the end of the while loop, 'start == end'
    return arr[start];
  }

  public static void main(String[] args) {
    Solution sol = new Solution();
    System.out.println(sol.findMax(new int[] { 1, 3, 8, 12, 4, 2 }));
    System.out.println(sol.findMax(new int[] { 3, 8, 3, 1 }));
```

You must run your code first

Time Complexity

Since we are reducing the search range by half at every step, this means that the time complexity of our algorithm will be O(logN) where 'N' is the total elements in the given array.

Space Complexity

The algorithm runs in constant space O(1).

← Previous
Bitonic Array Maximum (Easy)

Next →
Problem Challenge 1: Search Bitonic Array (Medium)

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